

Research Report & Scientific Writing

Presented by:

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The logo for Interpeko Research is displayed on a light beige background with a bokeh effect and orange curved borders. The word "Interpeko" is written in an orange script font with small decorative flourishes above the 'i' and 'o'. Below it, the word "Research" is written in a dark blue script font. A thin orange curved line is positioned under the word "Research".

Interpeko
Research

Learning Objectives

**By the End
of This
Session...**

*You will write
like a scientist*

- 1 *Define scientific writing and explain its role in evidence-based medicine.*
- 2 *Apply the IMRaD framework to structure a complete research paper.*
- 3 *Formulate a clinical research question using the PICO framework.*
- 4 *Apply the hierarchy of evidence to critically evaluate study quality.*
- 5 *Write a structured abstract, rigorous methods and unbiased results section.*
- 6 *Apply correct Vancouver referencing and use citation management tools.*
- 7 *Identify plagiarism, AI misuse and common errors that cause rejection.*

What is Scientific Writing?

Definition

Scientific writing is a formal, precise, and objective method of communicating research findings, methods, and conclusions to a specialist audience using evidence and structured argument.

Why Does It Matter?

Advances Knowledge

Research without publication is research lost. Scientific writing is how discoveries enter the global knowledge base and influence clinical practice worldwide.

Enables Peer Scrutiny

Written reports allow independent verification. Peer review checks for errors, bias, and invalid conclusions before findings reach clinicians or policymakers.

Drives Evidence-Based Medicine

Clinicians depend on published evidence to make treatment decisions. Poor writing leads to misinterpretation — which can directly harm patients.

The Writing Crisis in Medical Research

21%

*of researchers cite
writing quality as
their biggest
challenge*

50%

of papers rejected
due to poor writing — not poor science

3×

*higher acceptance
rate
for clearly written
manuscripts*

Source: Hess DR, Respir Care 2004 · Annesley TM, Clin Chem 2010

The 5 Pillars of Scientific Writing

CLARITY

1

Use simple,
direct
language.
One idea per
sentence.
Avoid jargon
unless
necessary.

ACCURACY

2

Every fact
must be
verifiable.
Cite your
sources.
No
overstatemen
t of findings.

CONCISENESS

3

Say what
needs to be
said.
Remove
redundant
words.
Value the
reader's time.

OBJECTIVITY

4

Report data
without bias.
Avoid
emotional
language.
Let evidence
speak for
itself.

INTEGRITY

5

No plagiarism.
No data
fabrication.
Full
attribution of
all sources.

Types of Scientific Documents — Part 1

Original Research Article

Reports entirely new primary research findings.

Must follow the IMRaD structure.

Subjected to full peer review before publication.

Highest individual academic value.

Systematic Review & Meta-Analysis

Exhaustively searches ALL published evidence on a topic.

Follows PRISMA 2020 reporting guidelines.

Meta-analysis pools data statistically if appropriate.

Represents the highest level in the evidence hierarchy.

Case Report & Case Series

Describes an unusual or instructive clinical presentation.

Follows CARE (CAse REport) guidelines.

Essential for documenting rare conditions and adverse events.

Requires explicit patient informed consent.

Types of Scientific Documents — Part 2

Review Article (Narrative Review)

Summarises a broad research area without formal meta-analysis.

Written by domain experts with a thematic perspective.

More flexible than systematic reviews.

Useful for introducing medical students to a topic.

Research Proposal

Outlines a planned study before execution begins.

Required for ethics committee approval and external funding.

Includes: background, objectives, methods, timeline & budget.

Precedes every formal research study.

Thesis & Dissertation

Extended scientific document produced for an academic degree.

Covers the entire research process in great depth.

Includes extended literature review and original findings.

Evaluated by both internal and external academic examiners.

What Makes a Good Research Question?

The FINER Criteria

F

Feasible: *Achievable within your available time, budget, sample size, and expertise. An unfeasible question wastes resources and yields no publishable results.*

I

Interesting: *The question must matter to the scientific or clinical community. It should spark genuine intellectual curiosity and motivate sustained research effort.*

N

Novel: *It must fill a genuine knowledge gap. Check existing literature systematically on PubMed, Cochrane, and EMBASE before finalising your question.*

E

Ethical: *The study must be approvable by an ethics committee. Participant safety, privacy, and dignity must be protected throughout the entire research process.*

R

Relevant: *The question must have meaningful impact on clinical practice, public health, or scientific theory. It should justify the time and cost of conducting the study.*

PICO Framework — Structuring Your Question

P Patient / Population

Who is your study population?
Adults aged 18–65 diagnosed with type 2 diabetes

I Intervention / Exposure

What is being studied?
Metformin 500 mg orally twice daily for 12 weeks

C Comparison / Control

What is the control or alternative?
Matched placebo tablets, same frequency and duration

O Outcome

What are you measuring?
Reduction in HbA1c (%) from baseline to 12 weeks

PICO — Complete Clinical Example

The Research Question:

"In adults with type 2 diabetes (P), does Metformin 500 mg twice daily (I) compared to placebo (C) significantly reduce HbA1c levels after 12 weeks of treatment (O)?"

Why PICO Matters

Keeps the question focused

Prevents scope creep and ensures your study has a single clear purpose.

Facilitates literature search

Use each PICO element as a search term to identify relevant prior evidence.

Guides study design

Each element of PICO directly maps to a component of your methods section.

Forms the hypothesis

The null hypothesis directly emerges from PICO: 'There is no difference in HbA1c...'

Hierarchy of Evidence — The Evidence Pyramid

Not all evidence is equal. Study design determines the strength of conclusions.

Systematic Reviews & Meta-Analyses

Highest — synthesises ALL RCTs on a topic

Randomised Controlled Trials (RCT)

Gold standard — randomisation removes selection bias

Cohort Studies

Prospective — establishes temporality

Case-Control Studies

Retrospective — good for rare diseases

Cross-Sectional Studies

Measures prevalence at one time point

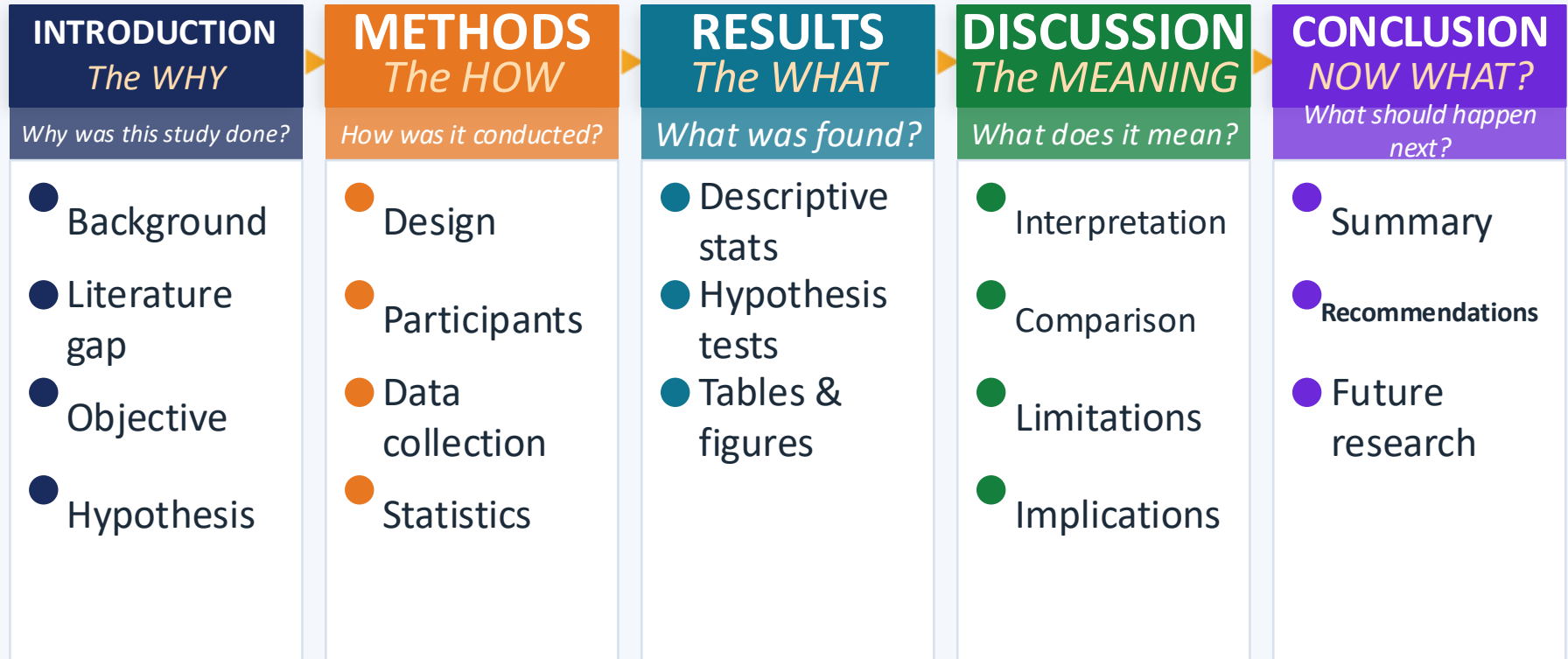
▼ **Lowest**

*Case Reports & Expert Opinion
Lowest — anecdotal, no control group*

**Publication quality: NEJM, Lancet, JAMA preferentially publish RCTs and systematic reviews.
Evidence level influences which journals will consider your manuscript.**

The IMRaD Framework — Overview

IMRaD = Introduction · Methods · Results · (and) · Discussion



IMRaD — The Logic Behind the Structure

Introduction = Funnel

Begin broad: establish global context and epidemiology.

Narrow to what is known and what is unknown.

End specific: state your exact objective and hypothesis.

Move from the general world → to your specific study.

Methods = Most Detailed Section

This must be reproducible — another researcher must be able to replicate your exact study.

Use past tense throughout (e.g., 'Data were collected...').

Subheadings are required: Design, Participants, Measurements, Analysis.

Never omit ethical approval information.

Results = Data Only — No Opinions

Present findings in a logical, pre-specified order.

Begin with Table 1: participant demographics. Report all pre-specified outcomes — not only significant ones.

Use past tense. Let the numbers speak.

Discussion = Funnel Reversed

Begin specific: restate your main finding first.

Broaden outward: compare with existing literature.

Acknowledge limitations honestly — this builds credibility.

End broad: state implications for practice and future research.

Crafting the Title

The title is the most-read part of any paper. It determines whether someone reads further.

12–15 Words

Short enough to be memorable. Long enough to convey the study's population, intervention, and design.

No Abbreviations

Spell everything out in full. Abbreviations reduce discoverability in database searches (e.g., PubMed, Scopus).

State the Study Design

Include the design when appropriate: '...A Randomised Controlled Trial' or '...A Systematic Review'.

Informative — Not Clever

Accurately reflects content. Avoid sensational language. Editors reject titles that overpromise findings.

Writing a Structured Abstract

Maximum 250–300 words · 5 mandatory sections · Must stand alone without the full paper

- 1 Background** State the problem and why it matters. 1–2 sentences maximum.
- 2 Objective** State the exact aim of the study. Begin with: 'The aim of this study was to...'
- 3 Methods** Design, sample size, setting, data collection tools, and statistical methods — in brief.
- 4 Results** Report key quantitative findings. Include exact statistics: p-values and 95% CI.
- 5 Conclusion** One sentence summary of the main finding and its clinical implication.

Keywords: Use 3–6 MeSH (Medical Subject Headings) terms. These drive discoverability on PubMed and Scopus.

Writing the Introduction — The Funnel

Think: Start WIDE → narrow to YOUR specific question

1

BROAD CONTEXT

Establish the global significance of the topic. Cite current epidemiology, disease burden, mortality, or economic cost. (2–3 sentences)

2

WHAT IS ALREADY KNOWN

Synthesise key evidence thematically — do NOT list papers chronologically. Group studies by finding, not by date. (3–4 sentences)

3

THE KNOWLEDGE GAP

'However...' 'Despite this...' 'To date, no study has evaluated...' Explicitly name what remains unknown. (1–2 sentences)

4

RATIONALE FOR THIS STUDY

Explain why your study is needed now. Link the gap directly to your chosen approach and population. (1–2 sentences)

5

OBJECTIVE + HYPOTHESIS

'The aim of this study was to...' State the null hypothesis (H_0) and alternative hypothesis (H_1) explicitly.

Methods — Part 1: Study Design & Participants

Golden Rule: A competent researcher must be able to exactly replicate your study from this section alone.

Study Design

State the design explicitly and upfront.

'A prospective, randomised, double-blind, placebo-controlled trial...'

Never assume the reader will infer the design from other clues.

Setting & Time Period

Name the institution (hospital, clinic, community).

State the exact study period: 'January 2023 to June 2024'.

Specify whether single-centre or multicentre.

Eligibility Criteria

List INCLUSION criteria: who qualified to participate.

List EXCLUSION criteria: who was excluded and why.

Both are essential — never omit either.

Sample Size & Power

State the calculated sample size with justification.

Report: expected effect size, power ($\geq 80\%$), alpha (0.05).

*Use G*Power or a published formula — cite your source.*

Methods — Part 2: Tools, Ethics & Statistics

Data Collection Tools

Name every instrument used. State whether it is validated — and cite the validation study.

Example: 'Depression was measured using the PHQ-9 (Kroenke et al., 2001).'

Ethical Approval

*State the IRB/ethics committee name and approval reference number.
Confirm that written informed consent was obtained from all participants.
Reference the Declaration of Helsinki.*

Statistical Analysis

*Name the software and version used: 'SPSS v.26 (IBM, USA)' or 'R v.4.3.0'.
List every test used (t-test, chi-square, logistic regression).
State: 'Statistical significance was set at $p < 0.05$ (two-tailed).'*

Primary & Secondary Outcomes

*Define the primary outcome explicitly: what you are measuring and how.
List secondary outcomes separately.
Pre-specification of outcomes prevents selective reporting bias.*

Writing the Results Section

✓ ALWAYS DO

- ✓ *Begin with Table 1 — participant demographics and baseline characteristics*
- ✓ *Report exact p-values ($p = 0.023$) and 95% Confidence Intervals*
- ✓ *Follow the same order as your Methods section*
- ✓ *Reference every table and figure explicitly in the text*

✗ NEVER DO

- ✗ *Interpret data or speculate — that belongs in the Discussion*
- ✗ *Repeat all table data verbatim — highlight key findings only*
- ✗ *Report only significant results — report ALL pre-specified outcomes*
- ✗ *Add results not mentioned in the Methods section (cherry-picking)*

Writing the Discussion — 6-Step Structure

1 State Your Main Finding

Open with your single most important result.
Do NOT repeat numbers — say what they mean.

3 Explain the Mechanism

Offer a plausible biological or clinical explanation.
Based on known physiology, pharmacology, or evidence.

5 Clinical Implications

What should clinicians or policymakers do?
This section gives your paper real-world relevance.

2 Compare with Literature

Does your finding agree or disagree with prior studies?
Cite relevant papers. Explain discrepancies.

4 Strengths & Limitations

Be transparent — all studies have limitations.
Honest acknowledgment actually increases credibility.

6 Future Research Directions

What specific questions remain unanswered?
Suggest feasible, realistic next steps.

Writing the Conclusion

The conclusion is the last thing the reader reads — and the first thing they remember.

1 Keep It SHORT

Aim for 100–200 words maximum. 1–3 concise paragraphs. The conclusion is a summary — not a new discussion.

2 Answer Your Research Question

Directly address whether your objective was achieved.
'This study demonstrated that Metformin significantly reduced HbA1c compared to placebo ($p < 0.001$)...'

3 No New Data

Never introduce data, statistics, or references that were not already presented in the Results section. This is a strict rule in all journals.

4 One Clinical Recommendation

What should clinicians or policy makers do based on your findings? State this clearly in one sentence.

5 One Future Research Direction

Suggest one specific, realistic, and feasible next research question that follows logically from your findings.

References — Vancouver Citation Style

Vancouver style is the international standard in medicine and health sciences.

Examples — Vancouver Format

Journal Article

Smith AB, Jones CD, Brown EF. Effect of metformin on glycaemic control in type 2 diabetes: a randomised controlled trial. *J Clin Endocrinol Metab.* 2022;107(4):e1234–40.

Book

Harrison TR. *Harrison's Principles of Internal Medicine.* 21st ed. New York: McGraw-Hill Education; 2022. p. 450–455.

Website

World Health Organization. Diabetes [Internet]. Geneva: WHO; 2023 [cited 2024 Jan 10]. Available from: <https://www.who.int/news-room/fact-sheets/detail/diabetes>

Rules: Number in order of first appearance · ≤6 authors list all; >6 write first 3 + 'et al.' · Use [1] or superscript in-text

Reference Management Tools

Mendeley

Free, cloud-based PDF manager.
Auto-generates citations in Word.
Available on all platforms.
Owned by Elsevier — widely used in medicine.

Zotero

Free and open-source.
Browser plugin for one-click saving.
Supports group libraries for collaboration.
Excellent for thesis work.

EndNote

Gold standard in academic research.
Powerful search and deduplication tools.
Requires institutional license.
Best for large systematic review projects.

RefWorks

Cloud-based, accessible from any device.
Direct integration with Google Docs and Word.
Ideal for students in collaborative projects.

Academic Language — Useful Phrases by Section

Introduction

- *"Previous studies have demonstrated that..."*
- *"The aetiology of X remains poorly understood..."*
- *"The aim of this study was to determine..."*

Methods

- *"Ethical approval was obtained from... (Ref: ...)"*
- *"Participants were randomly allocated to..."*
- *"Statistical significance was set at $p < 0.05$ (two-tailed)."*

Results

- *"There was a statistically significant difference between groups..."*
- *"Table 1 summarises the baseline characteristics..."*
- *"No significant association was observed between X and Y ($p = 0.42$)."*

Discussion

- *"These findings are consistent with those reported by [Author]..."*
- *"A possible explanation for this finding is..."*
- *"A key limitation of this study is the relatively small sample size..."*

Hedging Language — Degrees of Scientific Certainty

Scientific writing does NOT claim absolute certainty. Always reflect the actual degree of evidence.

HIGH CERTAINTY

confirms · demonstrates · proves · shows

Use only when evidence is extremely strong (e.g., multiple consistent RCTs).

MODERATE

suggests · indicates · appears to · supports

The most common level in medical research. Use by default.

LOW CERTAINTY

may · might · could · possibly · perhaps

Use when evidence is preliminary or based on a small sample.

UNCERTAINTY

it is unclear whether · warrants further investigation

Use when the evidence is contradictory or genuinely insufficient.

Words to AVOID: obviously · clearly · very · a lot · basically · things · interesting (without context) · nice · really

Statistical Reporting — What Must Be Included

Never report $p = 0.05$ alone. Always include: statistic + CI + effect size + sample size.

Sample Size & Power

Justify using a priori power analysis.

Target: Power $\geq 80\%$, Alpha = 0.05.

Cite the formula or software used (G*Power).

Descriptive Statistics

Normally distributed data $\rightarrow$ Mean $\pm$ SD.

Skewed or non-normal data $\rightarrow$ Median (IQR).

Always state which was used and why.

p-values & Confidence Intervals

Report exact p-values: $p = 0.023$ (not ' $p < 0.05$ ').

95% CI for all key estimates is mandatory.

CI shows precision — p-value alone does not.

Effect Size

Cohen's d · Odds Ratio (OR) · Risk Ratio (RR)

Number Needed to Treat (NNT) · Hazard Ratio (HR)

Effect size quantifies clinical importance.

Common Statistical Misinterpretations



p-value $\neq$ Clinical Significance

A $p = 0.04$ result may have no real clinical relevance. Always interpret alongside effect size and CI.
Example: A drug reduces blood pressure by 1 mmHg ($p = 0.03$) — statistically significant but clinically meaningless.



Correlation $\neq$ Causation

Two variables may correlate by chance, due to confounding, or reverse causality.
Observational studies CANNOT establish causation. Only RCTs can — with appropriate design.



Overfitting in Small Samples

Running many statistical tests on a small dataset increases false-positive results (Type I error).
Solution: Apply Bonferroni correction or pre-register your analysis plan before data collection.



Selective Reporting Bias

Reporting only statistically significant outcomes while omitting non-significant ones distorts the scientific record.
This is research misconduct. Pre-registration on ClinicalTrials.gov prevents this.

Top 10 Writing Errors That Cause Rejection

1 Abstract doesn't match the paper content

3 Methods section too vague to be replicated

5 Discussion fails to compare with existing literature

7 Inconsistent data between text, tables, and figures

9 References missing, incorrect, or wrongly formatted

2 Introduction lacks a clearly stated research gap

4 Results mix findings with interpretation

6 Missing p-values, confidence intervals, or effect sizes

8 Limitations not acknowledged or grossly understated

10 Grammatical errors and inconsistent use of tenses

Plagiarism — Types & Prevention

Plagiarism is the single most damaging form of research misconduct. It ends careers.

Direct Plagiarism

Copying another author's text word-for-word without quotation marks or citation.

The most severe form. Leads to immediate retraction and institutional sanctions.

Paraphrase Plagiarism

Rewording another author's idea slightly without proper attribution.

Changing a few words does not make the idea yours. Always cite the original source.

Self-Plagiarism

Reusing your own previously published work without disclosing or citing it.

Journals consider submitted work to be original. Prior publication must always be declared.

Mosaic Plagiarism

Mixing phrases from multiple sources without quotation marks or attribution.

The combination of sources does not make the text original — all must be cited.

Detection: Turnitin · iThenticate · Acceptable similarity: < 15–20% · Ghost writing is also research misconduct

AI in Scientific Writing — Acceptable Uses

AI tools (ChatGPT, Claude, Gemini) are assistants — NOT authors. Use them ethically and transparently.

Grammar & Language Editing

Polishing the language of a draft you have already written is acceptable.
AI corrects grammar, improves readability, and suggests clearer phrasing.
Your original ideas and content must remain entirely your own.

Paraphrasing Suggestions

AI can suggest alternative ways to express your existing ideas more clearly.
You must verify all suggestions for accuracy and appropriate scientific tone.

Literature Search Assistance

AI can suggest initial search terms and help you organise sources.
Always verify every reference manually on PubMed — AI cannot replace this step.

Formatting Assistance

AI can format tables, structure reference lists, and check word counts.
Useful for ensuring compliance with journal-specific author guidelines.

AI in Scientific Writing — Risks & Disclosure

Hallucinated References

AI generates fake but plausible-looking citations with real author names and journal titles. Every reference generated by AI MUST be verified manually on PubMed before use.

Fabricated Statistics & Data

AI may invent patient outcomes, trial results, or laboratory values that do not exist. Never use AI-generated data in any section of a scientific manuscript — this is fraud.

AI Cannot Be Listed as an Author

Per ICMJE 2023 guidelines, AI cannot fulfil the four authorship criteria. You remain fully responsible and accountable for all AI-assisted content you submit.

Disclosure Is Mandatory

NEJM, Lancet, BMJ, and Nature all require explicit disclosure of AI tool use in the Methods or Acknowledgments section of your manuscript. Non-disclosure constitutes academic misconduct at most institutions.

Authorship Criteria — ICMJE 2023 Standards

ALL four criteria must be met to qualify as an author:

1 Criterion 1: Substantial Contribution

To conception and design of the work, OR to acquisition, analysis, or interpretation of data.

2 Criterion 2: Drafting OR Critical Revision

Drafting the manuscript, OR critically revising it for important intellectual content.

3 Criterion 3: Final Approval

Final approval of the version to be published by all listed authors.

4 Criterion 4: Full Accountability

Agreement to be accountable for all aspects of the work, including data integrity and accuracy.

⚠ Gift & Ghost Authorship — Both Are Unethical

GIFT: Adding someone who did not contribute meaningfully (e.g., a department head) to honour or please them.
GHOST: Omitting someone who contributed substantially — often pharmaceutical company employees in clinical trials.

Journal Selection Strategy — Part 1

1 Scope Compatibility

Read the journal's published 'Aims & Scope' carefully.
Your topic, study design, and population must align precisely.
Submitting out-of-scope is the fastest route to immediate desk rejection.

2 Indexing Status

Target journals indexed in PubMed/MEDLINE, Scopus, or Web of Science.
Indexing = visibility = academic credibility.
Non-indexed journals carry little weight in promotion or academic assessment.

3 Impact Factor

Higher Impact Factor = greater prestige and wider readership.
For a 3rd-year student: IF 1–4 is realistic and still respected.
Don't over-aim — submitting to NEJM when you have a small cross-sectional study leads to rejection.

Journal Selection — Predatory Journals & Final Checks

4

Avoid Predatory Journals

Predatory journals charge publication fees but skip proper peer review.

They accept almost everything — which destroys your scientific credibility.

Check: Beall's List (beallslist.net) · Directory of Open Access Journals (DOAJ) · COPE membership.

5

Read the Author Guidelines

Every journal has specific requirements: word limits, reference formats, figure resolution, file types.

Non-compliance causes immediate rejection — even if the science is excellent.

Always read guidelines BEFORE writing your final draft.

6

Open Access vs. Subscription

Open Access: freely readable globally but requires Article Processing Charge (APC) — often \$1,000–5,000.

Subscription: free to publish but readers may need institutional access.

Many funding bodies (NIH, Wellcome) mandate open access for funded research.

The Peer Review Process

Journey of a manuscript from submission to publication



How to Respond to Reviewer Comments

- Address EVERY comment — ignoring any single comment leads to re-rejection
- Use a structured table: Reviewer Comment | Your Response | Manuscript Page/Line
- If you disagree — provide clear scientific justification politely and respectfully
- Highlight all text changes in the revised manuscript using track changes

Practical Example — Before Revision

✗ WEAK WRITING — Identify the Problems

"We did a study on diabetes patients. A lot of them got better with the new drug. The results were very interesting and clearly show the drug is good. We tested many patients and the data was basically positive. This is really important and clinicians should use this drug."

Problems Identified:

- **'A lot'** — No sample size or percentage stated — vague and unscientific
- **'Basically positive'** — No statistics — no p-value, no CI, no effect size whatsoever
- **'Very interesting'** — Informal language — 'very', 'interesting', 'good' are not scientific terms
- **'Clearly show'** — Overstates certainty — uses absolute language without justification
- **No study design** — What type of study was this? RCT? Cohort? Cross-sectional? Not stated.

Practical Example — After Revision

✓ STRONG SCIENTIFIC WRITING — See the Transformation

"In this randomised, double-blind, placebo-controlled trial, 124 adults with type 2 diabetes (mean age 54.2 ± 8.3 years) were allocated to receive Metformin 500 mg twice daily ($n = 62$) or matched placebo ($n = 62$) for 12 weeks. Metformin was associated with a statistically significant reduction in HbA1c from baseline compared to placebo (mean difference: -0.8% ; 95% CI: -1.2 to -0.4 ; $p < 0.001$; Cohen's $d = 0.68$)."

What Improved:

- **Study design explicitly stated** — Randomised, double-blind, placebo-controlled trial
- **Exact sample size given** — 124 patients with clear group allocation ($n = 62$ each)
- **Full statistics reported** — $p < 0.001$ + 95% CI (-1.2 to -0.4) + Cohen's $d = 0.68$
- **Hedged language used correctly** — 'was associated with' — not 'proves' or 'clearly shows'
- **Drug, dose & duration specified** — Metformin 500 mg twice daily $\times$ 12 weeks

Reporting Guidelines — Part 1

Mandatory checklists that ensure complete and transparent scientific reporting

CONSORT 2010

Randomised Controlled Trials

25-item checklist + mandatory participant flow diagram.
Required by NEJM, Lancet, JAMA, BMJ before submission.
Download: www.consort-statement.org

STROBE

Observational Studies

22-item checklist for cohort, case-control, and cross-sectional studies.
Published simultaneously in BMJ, Lancet, PLoS Med, and Annals of Internal Medicine.
Download: www.strobe-statement.org

PRISMA 2020

Systematic Reviews & Meta-Analyses

27-item checklist + mandatory PRISMA flow diagram showing all screening stages.
Must be completed and submitted alongside the manuscript.
Download: www.prisma-statement.org

CARE

Case Reports

13-item checklist covering timeline, clinical findings, and patient perspective.
Informed consent from the patient must be documented.
Download: www.care-statement.org

Reporting Guidelines — Part 2

STARD

Diagnostic Accuracy Studies

30-item checklist + patient flow diagram for sensitivity/specificity reporting.
Essential for studies validating diagnostic tests, biomarkers, or clinical prediction rules.
Download: www.stard-statement.org

SPIRIT

Clinical Trial Protocols

33-item checklist required before trial commencement.
Must be submitted alongside ethics application for all prospective clinical trials.
Download: www.spirit-statement.org

AGREE II

Clinical Practice Guidelines

23 items across 6 domains: Scope, Stakeholders, Rigour, Presentation, Applicability, Independence.
Used to rate the quality of any clinical practice guideline.
Download: www.agreetrust.org

EQUATOR

All Study Types — Umbrella
Resource

Central database for ALL reporting guidelines: www.equator-network.org
Always check here FIRST before beginning your manuscript.
Searchable by study type — provides links to all checklists and templates.



Always visit the EQUATOR Network (equator-network.org) to find the correct reporting guideline for your specific study type BEFORE submitting your manuscript.

3 Take-Messages

01

Structure Is Everything

Master IMRaD. Every section has exactly one purpose — never mix them.
Introduction = WHY · Methods = HOW · Results = WHAT · Discussion = WHAT IT MEANS

02

Precision Over Passion

Write with data — not with emotion. Report exact numbers, use hedging language, and always let your evidence speak clearly for itself.

03

Integrity Is Non-Negotiable

Cite everything. Follow ICMJE authorship criteria. Declare AI use.
One act of misconduct can permanently end a scientific career.

Conclusions



Scientific writing is a learnable skill — mastered through deliberate, consistent practice.



The IMRaD framework provides a universal, logical structure for all medical papers.



Each section has ONE purpose — Introduction explains WHY, Methods explains HOW.



PICO and the evidence hierarchy are the cornerstones of good research design.



Statistical accuracy, hedging language, and correct referencing are non-negotiable.



Follow ICMJE authorship criteria and report AI use transparently in every paper.



Use reporting guidelines — CONSORT, STROBE, PRISMA — for every study type you write.



References — Vancouver Format

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Thank You

Research Report & Scientific Writing

Dr. Mustafa

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Questions & Discussion Welcome

